

Basis SMEFTsim_topU31 (EFT SMEFT)

Basis used in the SMEFTsim_topU31 UFO models, version 3.0.0 or later. Implements Warsaw basis with $U(2)^3$ flavor symmetry in the quarks sector and $U(3)^2$ in the leptons sector. Q, t, b are left- and right-handed 3rd gen quarks, q, u, d are the left- and right-handed quark fields containing only the first two generations, and transforming as $U(2)$ -flavor doublets. ℓ, e are left- and right-handed lepton fields. Y_u, Y_d are the 2x2 Yukawas of up and down quarks in the first two generations. Y_l is the 3x3 lepton Yukawa. Yukawas defined by $L_{SM} \supset \bar{d}Y_d H^\dagger q$ and analogously for the others. Spurions connecting the first two generations with the 3rd are absent. In the UFO models, both Y_u and Y_d are assumed diagonal at the scale of evaluation, and the CKM is taken to be the unit matrix. Flavor indices are indicated with p, r, s, t with Einstein conventions on repeated indices. They run over 1,2 for quarks and 1,2,3 for leptons. This basis definition corresponds to a fixed `LambdaSMEFT=1e+3` in the UFO models. Notation and conventions can vary compared to the Warsaw basis paper, see arXiv:2012.11343 for all definitions.

Sectors

The effective Lagrangian is defined as

$$\mathcal{L}_{\text{eff}} = -\mathcal{H}_{\text{eff}} = \sum_{O_i=O_i^\dagger} C_i O_i + \sum_{O_i \neq O_i^\dagger} (C_i O_i + C_i^* O_i^\dagger).$$

`dB=de=dmu=dtau=0`

WC name	Operator	Type
cG	$f^{ABC} G_\mu^{A\nu} G_\nu^{B\rho} G_\rho^{C\mu} / \text{TeV}^2$	R
cGtil	$f^{ABC} \tilde{G}_\mu^{A\nu} G_\nu^{B\rho} G_\rho^{C\mu} / \text{TeV}^2$	R
cW	$\varepsilon^{IJK} W_\mu^{I\nu} W_\nu^{J\rho} W_\rho^{K\mu} / \text{TeV}^2$	R
cWtil	$\varepsilon^{IJK} \tilde{W}_\mu^{I\nu} W_\nu^{J\rho} W_\rho^{K\mu} / \text{TeV}^2$	R
cH	$(H^\dagger H)^3 / \text{TeV}^2$	R
cHbox	$(H^\dagger H) \square (H^\dagger H) / \text{TeV}^2$	R
cHDD	$(D_\mu H^\dagger H) (H^\dagger D^\mu H) / \text{TeV}^2$	R
cHG	$G_{\mu\nu}^A G^{A\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHGtil	$\tilde{G}_{\mu\nu}^A G^{A\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHW	$W_{\mu\nu}^I W^{I\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHWtil	$\tilde{W}_{\mu\nu}^I W^{I\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHB	$B_{\mu\nu} B^{\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHBtil	$\tilde{B}_{\mu\nu} B^{\mu\nu} H^\dagger H / \text{TeV}^2$	R
cHWB	$B_{\mu\nu} W^{I\mu\nu} H^\dagger \sigma^I H / \text{TeV}^2$	R
cHWBtil	$B_{\mu\nu} \tilde{W}^{I\mu\nu} H^\dagger \sigma^I H / \text{TeV}^2$	R
ceHRe	$(Y_l^\dagger)_{pr} (\bar{\ell}_p H e_r) (H^\dagger H) / \text{TeV}^2 + \text{h.c.}$	R

WC name	Operator	Type
ceHIm	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p H e_r)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cuHRe	$(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} u_r)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cuHIm	$i(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} u_r)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
ctHRe	$(\bar{Q} \tilde{H} t)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
ctHIm	$i(\bar{Q} \tilde{H} t)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cdHRe	$(Y_d^\dagger)_{pr}(\bar{q}_p H d_r)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cdHIm	$i(Y_d^\dagger)_{pr}(\bar{q}_p H d_r)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cbHRe	$(\bar{Q} H b)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
cbHIm	$i(\bar{Q} H b)(H^\dagger H)/\text{TeV}^2 + \text{h.c.}$	R
ceWRe	$(Y_l^\dagger)_{pr}(\bar{\ell}_p \sigma^I H \sigma^{\mu\nu} e_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
ceWIm	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p \sigma^I H \sigma^{\mu\nu} e_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
ceBRe	$(Y_l^\dagger)_{pr}(\bar{\ell}_p H \sigma^{\mu\nu} e_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
ceBIm	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p H \sigma^{\mu\nu} e_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cuGRe	$(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} \sigma^{\mu\nu} T^A u_r) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cuGIm	$i(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} \sigma^{\mu\nu} T^A u_r) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
ctGRe	$(\bar{Q} \tilde{H} \sigma^{\mu\nu} T^A t) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
ctGIm	$i(\bar{Q} \tilde{H} \sigma^{\mu\nu} T^A t) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cuWRe	$(Y_u^\dagger)_{pr}(\bar{q}_p \sigma^I \tilde{H} \sigma^{\mu\nu} u_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cuWIm	$i(Y_u^\dagger)_{pr}(\bar{q}_p \sigma^I \tilde{H} \sigma^{\mu\nu} u_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
ctWRe	$(\bar{Q} \sigma^I \tilde{H} \sigma^{\mu\nu} t) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
ctWIm	$i(\bar{Q} \sigma^I \tilde{H} \sigma^{\mu\nu} t) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cuBRe	$(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} \sigma^{\mu\nu} u_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cuBIm	$i(Y_u^\dagger)_{pr}(\bar{q}_p \tilde{H} \sigma^{\mu\nu} u_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
ctBRe	$(\bar{Q} \tilde{H} \sigma^{\mu\nu} t) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
ctBIm	$i(\bar{Q} \tilde{H} \sigma^{\mu\nu} t) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cdGRe	$(Y_d^\dagger)_{pr}(\bar{q}_p H \sigma^{\mu\nu} T^A d_r) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cdGIm	$i(Y_d^\dagger)_{pr}(\bar{q}_p H \sigma^{\mu\nu} T^A d_r) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cbGRe	$(\bar{Q} H \sigma^{\mu\nu} T^A b) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cbGIm	$i(\bar{Q} H \sigma^{\mu\nu} T^A b) G_{\mu\nu}^A / \text{TeV}^2 + \text{h.c.}$	R
cdWRe	$(Y_d^\dagger)_{pr}(\bar{q}_p \sigma^I H \sigma^{\mu\nu} d_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cdWIm	$i(Y_d^\dagger)_{pr}(\bar{q}_p \sigma^I H \sigma^{\mu\nu} d_r) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cbWRe	$(\bar{Q} \sigma^I H \sigma^{\mu\nu} b) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cbWIm	$i(\bar{Q} \sigma^I H \sigma^{\mu\nu} b) W_{\mu\nu}^I / \text{TeV}^2 + \text{h.c.}$	R
cdBRe	$(Y_d^\dagger)_{pr}(\bar{q}_p H \sigma^{\mu\nu} d_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cdBIm	$i(Y_d^\dagger)_{pr}(\bar{q}_p H \sigma^{\mu\nu} d_r) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cbBRe	$(\bar{Q} H \sigma^{\mu\nu} b) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
cbBIm	$i(\bar{Q} H \sigma^{\mu\nu} b) B_{\mu\nu} / \text{TeV}^2 + \text{h.c.}$	R
ch11	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{\ell}_p \gamma^\mu \ell_p) / \text{TeV}^2$	R
ch13	$(H^\dagger i \overleftrightarrow{D}_\mu^I H)(\bar{\ell}_p \gamma^\mu \sigma^I \ell_p) / \text{TeV}^2$	R
chj1	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{q}_p \gamma^\mu q_p) / \text{TeV}^2$	R

WC name	Operator	Type
cHj3	$(H^\dagger i \overleftrightarrow{D}_\mu^I H)(\bar{q}_p \gamma^\mu \sigma^I q_p)/\text{TeV}^2$	R
cHQ1	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{Q} \gamma^\mu Q)/\text{TeV}^2$	R
cHQ3	$(H^\dagger i \overleftrightarrow{D}_\mu^I H)(\bar{Q} \gamma^\mu \sigma^I Q)/\text{TeV}^2$	R
cHe	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{e}_p \gamma^\mu e_p)/\text{TeV}^2$	R
cHu	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{u}_p \gamma^\mu u_p)/\text{TeV}^2$	R
cHt	$(H^\dagger i \overleftrightarrow{D}_\mu^I H)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R
cHd	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{d}_p \gamma^\mu d_p)/\text{TeV}^2$	R
cHbq	$(H^\dagger i \overleftrightarrow{D}_\mu H)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cHudRe	$(Y_u Y_d^\dagger)_{pr}(\tilde{H}^\dagger i D_\mu H)(\bar{u}_p \gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R
cHudIm	$i(Y_u Y_d^\dagger)_{pr}(\tilde{H}^\dagger i D_\mu H)(\bar{u}_p \gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R
cHtbRe	$(\tilde{H}^\dagger i D_\mu H)(\bar{t} \gamma^\mu b)/\text{TeV}^2 + \text{h.c.}$	R
cHtbIm	$i(\tilde{H}^\dagger i D_\mu H)(\bar{t} \gamma^\mu b)/\text{TeV}^2 + \text{h.c.}$	R
c11	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{\ell}_r \gamma^\mu \ell_r)/\text{TeV}^2$	R
c111	$(\bar{\ell}_p \gamma_\mu \ell_r)(\bar{\ell}_r \gamma^\mu \ell_p)/\text{TeV}^2$	R
c1j1	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{q}_r \gamma^\mu q_r)/\text{TeV}^2$	R
c1j3	$(\bar{\ell}_p \gamma_\mu \sigma^I \ell_p)(\bar{q}_r \gamma^\mu \sigma^I q_r)/\text{TeV}^2$	R
cQ11	$(\bar{Q} \gamma_\mu Q)(\bar{\ell}_p \gamma^\mu \ell_p)/\text{TeV}^2$	R
cQ13	$(\bar{Q} \gamma_\mu \sigma^I Q)(\bar{\ell}_p \gamma^\mu \sigma^I \ell_p)/\text{TeV}^2$	R
cjj11	$(\bar{q}_p \gamma_\mu q_p)(\bar{q}_r \gamma^\mu q_r)/\text{TeV}^2$	R
cjj18	$(\bar{q}_p \gamma_\mu T^A q_p)(\bar{q}_r \gamma^\mu T^A q_r)/\text{TeV}^2$	R
cjj31	$(\bar{q}_p \gamma_\mu \sigma^I q_p)(\bar{q}_r \gamma^\mu \sigma^I q_r)/\text{TeV}^2$	R
cjj38	$(\bar{q}_p \gamma_\mu \sigma^I T^A q_p)(\bar{q}_r \gamma^\mu \sigma^I T^A q_r)/\text{TeV}^2$	R
cQQ1	$(\bar{Q} \gamma_\mu Q)(\bar{Q} \gamma^\mu Q)/\text{TeV}^2$	R
cQQ8	$(\bar{Q} \gamma_\mu T^A Q)(\bar{Q} \gamma^\mu T^A Q)/\text{TeV}^2$	R
cQj11	$(\bar{Q} \gamma_\mu Q)(\bar{q}_p \gamma^\mu q_p)/\text{TeV}^2$	R
cQj18	$(\bar{Q} \gamma_\mu T^A Q)(\bar{q}_p \gamma^\mu T^A q_p)/\text{TeV}^2$	R
cQj31	$(\bar{Q} \gamma_\mu \sigma^I Q)(\bar{q}_p \gamma^\mu \sigma^I q_p)/\text{TeV}^2$	R
cQj38	$(\bar{Q} \gamma_\mu \sigma^I T^A Q)(\bar{q}_p \gamma^\mu \sigma^I T^A q_p)/\text{TeV}^2$	R
cee	$(\bar{e}_p \gamma_\mu e_p)(\bar{e}_r \gamma^\mu e_r)/\text{TeV}^2$	R
cuu1	$(\bar{u}_p \gamma_\mu u_p)(\bar{u}_r \gamma^\mu u_r)/\text{TeV}^2$	R
cuu8	$(\bar{u}_p \gamma_\mu T^A u_p)(\bar{u}_r \gamma^\mu T^A u_r)/\text{TeV}^2$	R
ctt	$(\bar{t} \gamma_\mu t)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R
ctu1	$(\bar{t} \gamma_\mu t)(\bar{u}_p \gamma^\mu u_p)/\text{TeV}^2$	R
ctu8	$(\bar{t} \gamma_\mu T^A t)(\bar{u}_p \gamma^\mu T^A u_p)/\text{TeV}^2$	R
cdd1	$(\bar{d}_p \gamma_\mu d_p)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
cdd8	$(\bar{d}_p \gamma_\mu T^A d_p)(\bar{d}_r \gamma^\mu T^A d_r)/\text{TeV}^2$	R
cbb	$(\bar{b} \gamma_\mu b)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cbd1	$(\bar{b} \gamma_\mu b)(\bar{d}_p \gamma^\mu d_p)/\text{TeV}^2$	R
cbd8	$(\bar{b} \gamma_\mu T^A b)(\bar{d}_p \gamma^\mu T^A d_p)/\text{TeV}^2$	R
ceu	$(\bar{e}_p \gamma_\mu e_p)(\bar{u}_r \gamma^\mu u_r)/\text{TeV}^2$	R
cte	$(\bar{e}_p \gamma_\mu e_p)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R

WC name	Operator	Type
ced	$(\bar{e}_p \gamma_\mu e_p)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
cbe	$(\bar{e}_p \gamma_\mu e_p)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cud1	$(\bar{u}_p \gamma_\mu u_p)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
ctd1	$(\bar{t} \gamma_\mu t)(\bar{d}_p \gamma^\mu d_p)/\text{TeV}^2$	R
cbu1	$(\bar{u}_p \gamma_\mu u_p)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
ctb1	$(\bar{t} \gamma_\mu t)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cud8	$(\bar{u}_p \gamma_\mu T^A u_p)(\bar{d}_r \gamma^\mu T^A d_r)/\text{TeV}^2$	R
ctd8	$(\bar{t} \gamma_\mu T^A t)(\bar{d}_p \gamma^\mu T^A d_p)/\text{TeV}^2$	R
cbu8	$(\bar{u}_p \gamma_\mu T^A u_p)(\bar{b} \gamma^\mu T^A b)/\text{TeV}^2$	R
ctb8	$(\bar{t} \gamma_\mu T^A t)(\bar{b} \gamma^\mu T^A b)/\text{TeV}^2$	R
cutbd1Re	$(Y_u Y_d^\dagger)_{pr}(\bar{u}_p \gamma_\mu t)(\bar{b} \gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R
cutbd1Im	$i(Y_u Y_d^\dagger)_{pr}(\bar{u}_p \gamma_\mu t)(\bar{b} \gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R
cutbd8Re	$(Y_u Y_d^\dagger)_{pr}(\bar{u}_p \gamma_\mu T^A t)(\bar{b} \gamma^\mu T^A d_r)/\text{TeV}^2 + \text{h.c.}$	R
cutbd8Im	$i(Y_u Y_d^\dagger)_{pr}(\bar{u}_p \gamma_\mu T^A t)(\bar{b} \gamma^\mu T^A d_r)/\text{TeV}^2 + \text{h.c.}$	R
cle	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{e}_r \gamma^\mu e_r)/\text{TeV}^2$	R
clu	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{u}_r \gamma^\mu u_r)/\text{TeV}^2$	R
ctl	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R
cll	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
cb1	$(\bar{\ell}_p \gamma_\mu \ell_p)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cje	$(\bar{q}_p \gamma_\mu q_p)(\bar{e}_r \gamma^\mu e_r)/\text{TeV}^2$	R
cqe	$(\bar{Q} \gamma_\mu Q)(\bar{e}_r \gamma^\mu e_r)/\text{TeV}^2$	R
cju1	$(\bar{q}_p \gamma_\mu q_p)(\bar{u}_r \gamma^\mu u_r)/\text{TeV}^2$	R
cqu1	$(\bar{Q} \gamma_\mu Q)(\bar{u}_r \gamma^\mu u_r)/\text{TeV}^2$	R
ctj1	$(\bar{q}_p \gamma_\mu q_p)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R
cqt1	$(\bar{Q} \gamma_\mu Q)(\bar{t} \gamma^\mu t)/\text{TeV}^2$	R
cju8	$(\bar{q}_p \gamma_\mu T^A q_p)(\bar{u}_r \gamma^\mu T^A u_r)/\text{TeV}^2$	R
cqu8	$(\bar{Q} \gamma_\mu T^A Q)(\bar{u}_r \gamma^\mu T^A u_r)/\text{TeV}^2$	R
ctj8	$(\bar{q}_p \gamma_\mu T^A q_p)(\bar{t} \gamma^\mu T^A t)/\text{TeV}^2$	R
cqt8	$(\bar{Q} \gamma_\mu T^A Q)(\bar{t} \gamma^\mu T^A t)/\text{TeV}^2$	R
cjd1	$(\bar{q}_p \gamma_\mu q_p)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
cqd1	$(\bar{Q} \gamma_\mu Q)(\bar{d}_r \gamma^\mu d_r)/\text{TeV}^2$	R
cbj1	$(\bar{q}_p \gamma_\mu q_p)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cqb1	$(\bar{Q} \gamma_\mu Q)(\bar{b} \gamma^\mu b)/\text{TeV}^2$	R
cjd8	$(\bar{q}_p \gamma_\mu T^A q_p)(\bar{d}_r \gamma^\mu T^A d_r)/\text{TeV}^2$	R
cqd8	$(\bar{Q} \gamma_\mu T^A Q)(\bar{d}_r \gamma^\mu T^A d_r)/\text{TeV}^2$	R
cbj8	$(\bar{q}_p \gamma_\mu T^A q_p)(\bar{b} \gamma^\mu T^A b)/\text{TeV}^2$	R
cqb8	$(\bar{Q} \gamma_\mu T^A Q)(\bar{b} \gamma^\mu T^A b)/\text{TeV}^2$	R
cjqtu1Re	$(Y_u^\dagger)_{pr}(\bar{q}_p \gamma_\mu Q)(\bar{t} \gamma^\mu u_r)/\text{TeV}^2 + \text{h.c.}$	R
cjqtu1Im	$i(Y_u^\dagger)_{pr}(\bar{q}_p \gamma_\mu Q)(\bar{t} \gamma^\mu u_r)/\text{TeV}^2 + \text{h.c.}$	R
cjqtu8Re	$(Y_u^\dagger)_{pr}(\bar{q}_p \gamma_\mu T^A Q)(\bar{t} \gamma^\mu T^A u_r)/\text{TeV}^2 + \text{h.c.}$	R
cjqtu8Im	$i(Y_u^\dagger)_{pr}(\bar{q}_p \gamma_\mu T^A Q)(\bar{t} \gamma^\mu T^A u_r)/\text{TeV}^2 + \text{h.c.}$	R
cjqbd1Re	$(Y_d^\dagger)_{pr}(\bar{q}_p \gamma_\mu Q)(\bar{b} \gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R

WC name	Operator	Type
cjQbd1Im	$i(Y_d^\dagger)_{pr}(\bar{q}_p\gamma_\mu Q)(\bar{b}\gamma^\mu d_r)/\text{TeV}^2 + \text{h.c.}$	R
cjQbd8Re	$(Y_d^\dagger)_{pr}(\bar{q}_p\gamma_\mu T^A Q)(\bar{b}\gamma^\mu T^A d_r)/\text{TeV}^2 + \text{h.c.}$	R
cjQbd8Im	$i(Y_d^\dagger)_{pr}(\bar{q}_p\gamma_\mu T^A Q)(\bar{b}\gamma^\mu T^A d_r)/\text{TeV}^2 + \text{h.c.}$	R
cledjRe	$(Y_l^\dagger)_{pr}Y_{d,st}(\bar{\ell}_p^I e_r)(\bar{d}_s q_t^I)/\text{TeV}^2 + \text{h.c.}$	R
cledjIm	$i(Y_l^\dagger)_{pr}Y_{d,st}(\bar{\ell}_p^I e_r)(\bar{d}_s q_t^I)/\text{TeV}^2 + \text{h.c.}$	R
clebQRe	$(Y_l^\dagger)_{pr}(\bar{\ell}_p^I e_r)(\bar{b}Q^I)/\text{TeV}^2 + \text{h.c.}$	R
clebQIm	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p^I e_r)(\bar{b}Q^I)/\text{TeV}^2 + \text{h.c.}$	R
cjujd1Re	$(Y_u^\dagger)_{pr}(Y_d^\dagger)_{st}(\bar{q}_p^I u_r)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd1Im	$i(Y_u^\dagger)_{pr}(Y_d^\dagger)_{st}(\bar{q}_p^I u_r)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd11Re	$(Y_u^\dagger)_{sr}(Y_d^\dagger)_{pt}(\bar{q}_p^I u_r)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd11Im	$i(Y_u^\dagger)_{sr}(Y_d^\dagger)_{pt}(\bar{q}_p^I u_r)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtjd1Re	$(Y_d^\dagger)_{st}(\bar{Q}^I t)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtjd1Im	$i(Y_d^\dagger)_{st}(\bar{Q}^I t)(\bar{q}_s^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjuqb1Re	$(Y_u^\dagger)_{pr}(\bar{q}_p^I u_r)(\bar{Q}^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjuqb1Im	$i(Y_u^\dagger)_{pr}(\bar{q}_p^I u_r)(\bar{Q}^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQujb1Re	$(Y_u^\dagger)_{sr}(\bar{Q}^I u_r)(\bar{q}_s^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQujb1Im	$i(Y_u^\dagger)_{sr}(\bar{Q}^I u_r)(\bar{q}_s^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjtQd1Re	$(Y_d^\dagger)_{pt}(\bar{q}_p^I t)(\bar{Q}^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjtQd1Im	$i(Y_d^\dagger)_{pt}(\bar{q}_p^I t)(\bar{Q}^J d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtqb1Re	$(\bar{Q}^I t)(\bar{Q}^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtqb1Im	$i(\bar{Q}^I t)(\bar{Q}^J b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd8Re	$(Y_u^\dagger)_{pr}(Y_d^\dagger)_{st}(\bar{q}_p^I T^A u_r)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd8Im	$i(Y_u^\dagger)_{pr}(Y_d^\dagger)_{st}(\bar{q}_p^I T^A u_r)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd81Re	$(Y_u^\dagger)_{sr}(Y_d^\dagger)_{pt}(\bar{q}_p^I T^A u_r)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjujd81Im	$i(Y_u^\dagger)_{sr}(Y_d^\dagger)_{pt}(\bar{q}_p^I T^A u_r)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtjd8Re	$(Y_d^\dagger)_{st}(\bar{Q}^I T^A t)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtjd8Im	$i(Y_d^\dagger)_{st}(\bar{Q}^I T^A t)(\bar{q}_s^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjuqb8Re	$(Y_u^\dagger)_{pr}(\bar{q}_p^I T^A u_r)(\bar{Q}^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjuqb8Im	$i(Y_u^\dagger)_{pr}(\bar{q}_p^I T^A u_r)(\bar{Q}^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQujb8Re	$(Y_u^\dagger)_{sr}(\bar{Q}^I T^A u_r)(\bar{q}_s^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQujb8Im	$i(Y_u^\dagger)_{sr}(\bar{Q}^I T^A u_r)(\bar{q}_s^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjtQd8Re	$(Y_d^\dagger)_{pt}(\bar{q}_p^I T^A t)(\bar{Q}^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cjtQd8Im	$i(Y_d^\dagger)_{pt}(\bar{q}_p^I T^A t)(\bar{Q}^J T^A d_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtqb8Re	$(\bar{Q}^I T^A t)(\bar{Q}^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cQtqb8Im	$i(\bar{Q}^I T^A t)(\bar{Q}^J T^A b)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleju1Re	$(Y_l^\dagger)_{pr}(Y_u^\dagger)_{st}(\bar{\ell}_p^I e_r)(\bar{q}_s^J u_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleju1Im	$i(Y_l^\dagger)_{pr}(Y_u^\dagger)_{st}(\bar{\ell}_p^I e_r)(\bar{q}_s^J u_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleQt1Re	$(Y_l^\dagger)_{pr}(\bar{\ell}_p^I e_r)(\bar{Q}^J t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleQt1Im	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p^I e_r)(\bar{Q}^J t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleju3Re	$(Y_l^\dagger)_{pr}(Y_u^\dagger)_{st}(\bar{\ell}_p^I \sigma_{\mu\nu} e_r)(\bar{q}_s^J \sigma^{\mu\nu} u_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R

WC name	Operator	Type
cleju3Im	$i(Y_l^\dagger)_{pr}(Y_u^\dagger)_{st}(\bar{\ell}_p^I\sigma_{\mu\nu}e_r)(\bar{q}_s^J\sigma^{\mu\nu}u_t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleQt3Re	$(Y_l^\dagger)_{pr}(\bar{\ell}_p^I\sigma_{\mu\nu}e_r)(\bar{Q}^J\sigma^{\mu\nu}t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R
cleQt3Im	$i(Y_l^\dagger)_{pr}(\bar{\ell}_p^I\sigma_{\mu\nu}e_r)(\bar{Q}^J\sigma^{\mu\nu}t)\varepsilon_{IJ}/\text{TeV}^2 + \text{h.c.}$	R